

7-1 Hands on Project

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Forensic Tools for Mac Devices							
Tool	Price	Acquisition Methods	Key Features	Professional Tool / App	Pros	Cons	Best For
FTK Imager	Free	Disk imaging, file system analysis	Supports macOS file systems, creates bit-for-bit copies	Professional	Free, widely used in the industry	Limited macOS-specific features	Individual investigators, small firms
Autopsy	Free	Disk imaging, file system analysis	Open-source, supports multiple file systems, timeline analysis	Open-source	Free, extensible	Requires technical expertise	
RECON ITR / RECON LAB	\$1,199 / \$2,399	Live data acquisition, disk imaging, RAM imaging	Supports macOS 10.6+, timeline reports, thousands of artifacts	Professional	User-friendly interface, comprehensive artifact identification	Limited support for iOS devices	Small to medium-sized firms, individual investigators
Forensic Tools for iPod Devices							
Tool	Price	Acquisition Methods	Key Features	Professional Tool / App	Pros	Cons	Best For
iTunes Backup Extraction	Free	Accessing iTunes backups	Extracts data from iPod backups	App / Free	Free, straightforward	Limited to available backups, no direct device acquisition	Individual investigators, preliminary investigations
iRip	\$9.95	Media transfer, backup extraction, wide file range support	iTunes-style interface, supports all iPods and iPhones	App / Paid	Simple to use, integrates with iTunes	Limited forensic analysis capabilities	Individual investigators, media-focused investigations
iMazing	Around \$28 per device for a year coverage	Live data acquisition, backup extraction	File system access, app data extraction, backup management	Professional	User-friendly interface, supports multiple iOS devices	Limited forensic analysis capabilities	Investigators focusing on iOS data management
Elcomsoft iOS Forensic Toolkit	\$2,199	Full file system & logical extraction, checkm8	Keychain decryption, passcode bypass, supports legacy devices	Professional	Deep access to iOS systems, supports legacy devices	Primarily focused on iOS devices	Professional firms, team use

Forensic Tools for iPhone Devices							
Tool	Price	Acquisition Methods	Key Features	Professional Tool / App	Pros	Cons	Best For
Elcomsoft iOS Forensic Toolkit	\$2,199	Logical & physical extraction, checkm8	Keychain decryption, passcode bypass, supports legacy devices	Professional	Deep access to iOS systems, supports legacy devices	Primarily focused on iOS devices	Professional firms, team use
Magnet AXIOM	5K-8K annually	Logical, physical, cloud extraction	Multi-platform, app data, GPS, browser, cloud	Professional	Holistic evidence view, cross-platform support	High cost	Large firms, team use
Cellebrite UFED	36K-38K for 100 unlocks	Logical, physical, file system extraction	Wide device support, advanced decryption capabilities	Professional	Trusted, supports locked/damaged devices	Expensive	Large firms, multiple investigators
XRY	\$2,000-\$6,000 annually	Logical & physical extraction	Wide device support, advanced decryption capabilities	Professional	Trusted, supports locked/damaged devices	Expensive	Large firms, multiple investigators
Oxygen Forensic Detective	\$3,695 annually and \$9,495 one-time	Logical & physical extraction, cloud extraction	App data, cloud integration, GPS, browser	Professional	Comprehensive data extraction, supports multiple platforms	Expensive	Large firms, team use

In digital forensics, it is crucial that the right forensics tools are chosen to address the needs of the firm employing them. When conducting investigations at a small firm, unique challenges such as limited budgets, fewer staff, and the need to cover multiple device types are faced by investigators. Investigators must balance cost, ease of use, and functionality to ensure investigations are thorough without exceeding financial constraints. The above tables outline the tools chosen that support Mac computers, iPods, and iPhones, provide access to app data, backups, and file systems, and handle encrypted data when necessary. This paper will provide a detailed outline for which of these tools will best suit the needs of an investigation at a small firm and the challenges that will be faced.

In a small forensic firm, expensive forensic tools such as Cellebrite UFED or Magnet AXIOM can't be afforded. Instead, affordable or free options are preferred, especially when they are well-documented and easy to use. When searching for forensic tools, essential functions include disk imaging for Macs, logical and backup acquisition for iPods and iPhones, and the ability to extract all types of digital data. The search for a suitable tool for a small forensic firm demands user-friendliness and well-documentation as the top two priorities. There are usually not a lot of people working at small firms, which makes familiar tools an invaluable asset. In addition to those top two priorities, having a wide-range of capabilities and being budget friendly are also heavily prioritised.

For Mac devices, FTK Imager and Autopsy are free and capable of disk imaging and file system analysis, though FTK Imager requires more technical knowledge to properly disk image Mac drives. RECON ITR/RECON LAB offers live acquisition and memory imaging, but comes with higher costs. For iPod devices, iTunes Backup Extraction allows investigators to access backups for free, while iRip and iMazing provide additional capabilities such as live data

acquisition and app data extraction at modest prices. Elcomsoft iOS Forensic Toolkit offers advanced functions like keychain decryption and passcode bypass, suitable for legacy or encrypted devices. For iPhones, professional-grade tools such as Cellebrite UFED, Magnet AXIOM, XRY, and Oxygen Forensic Detective provide logical, physical, and cloud extraction, but their high cost makes them less practical for small firms.

Considering budget, usability, and functionality, the best choice for a small firm investigator would be FTK Imager for Macs, iTunes Backup Extraction and iMazing for iPods and iPhones, with use of Elcomsoft iOS Forensic Toolkit for iPhone cases. FTK Imager provides reliable disk imaging and file system analysis at no cost, while iTunes Backup Extraction allows investigators to access essential iPod data without paying for expensive software. iMazing adds flexibility with live acquisition and backup management across multiple iOS devices. Elcomsoft can be used on cases involving encrypted devices, situations requiring advanced keychain decryption, and legacy iPhones. This combination of software allows investigators at a small firm to complete their cases properly at a reasonable cost.

In summary, a small firm investigator must select tools that maximize coverage and functionality while staying within a budget. By combining free and moderately priced tools such as FTK Imager, iTunes Backup Extraction, iMazing, and Elcomsoft iOS Forensic Toolkit, a small firm could efficiently manage forensic investigations across Macs, iPods, and iPhones. This approach ensures reliable evidence acquisition without the prohibitive costs of expensive software.

